Muhammad Waqar

Islamabad, Pakistan | +92 308 6715563 | waqarsahi621@gmail.com
LinkedIn | Portfolio

Summary

AI Lead Engineer with 7+ years of experience taking machine learning and generative AI systems from prototype to production. Currently leads agentic AI and RAG development at Softoo, including an LLM-driven audit platform for SAMA, ISO and BCM compliance. Hands-on with multi-agent orchestration (LangGraph, LangChain), RAG, text-to-SQL, and hybrid deployments that pair on-premise open-source LLMs with Azure OpenAI for data-sensitive clients. Previously led ML teams delivering computer vision and NLP systems, and published at IEEE ICRAI.

Experience

Softoo May 2025 – Present
AI Lead Engineer (Generative AI, RAG Systems, AI Agents) *Islamabad, Pakistan*

- Lead agentic AI and RAG development for enterprise automation and decision-support platforms, owning the architecture from design through production.
- Architected multi-agent workflows with LangGraph and LangChain that plan and execute multi-step tasks through tool calling, replacing fixed, rule-based pipelines.
- Built RAG pipelines with vector databases and semantic search that ground LLM answers in enterprise knowledge bases and cite their sources.
- Deployed SQLCoder, Llama and Code Llama on-premise for clients whose data cannot leave their network, keeping data private and lowering inference cost compared with hosted APIs.
- Led development of the AI audit platform for BCM, SAMA and ISO compliance (see Projects), covering risk and control generation, control testing and audit reporting.
- Mentor junior engineers and set team practices for agent development and model governance in production.

Forbmax (Pvt) Ltd May 2023 – May 2025
AI Team Lead (Computer Vision, NLP, Data Science) *Islamabad, Pakistan*

- Led a team of machine learning engineers across computer vision, NLP and data science, setting technical direction and owning core system architecture.
- Rebuilt end-to-end model pipelines (preprocessing, training, deployment) around automated workflows, improving operational efficiency by 40%.
- Fine-tuned transformer, YOLO and vision transformer models with transfer learning to reach target accuracy on client-specific datasets.
- Built an active-learning annotation system that reduced manual labelling effort by 70% while keeping label quality consistent.
- Moved model delivery from one-off scripts to APIs and microservices, so models integrate cleanly with external products and platforms.

Horizon Tech Services Jun 2020 – Apr 2023
Senior AI Engineer *Islamabad, Pakistan*

- Owned ML projects end to end (data collection, preprocessing, training, deployment) and mentored a team of engineers through each stage.
- Built preprocessing pipelines that automated feature extraction and data augmentation, improving model performance and cutting manual data work.
- Led fine-tuning of object detection and tracking models for production applications, improving recognition and tracking accuracy.

Lampro Mellon IT Solutions (Pvt) Ltd Aug 2019 – May 2020
Python Developer *Lahore, Pakistan*

- Developed Python software components with a focus on verification and high-level design.
- Implemented algorithms and data-processing modules, and wrote automated verification scripts to replace manual testing.

Projects

Enterprise Compliance AI Agent Platform (Softoo) | *Agentic AI, RAG, LangGraph* Sep 2026

- Led the design of an agentic AI audit platform for BCM, SAMA and ISO compliance, including automated maturity assessments across policies, contracts and internal documentation.
- Built the end-to-end audit workflow with LangGraph and LangChain: engagement planning, risk and control matrix generation, control testing and audit-ready reports.
- Built a RAG pipeline with vector stores and semantic search that grounds LLM outputs in enterprise documents, improving factual accuracy and traceability.
- Combined self-hosted models (Llama, DeepSeek) for sensitive data with Azure OpenAI/OpenAI for generation where client policy allows.
- Added guardrails, audit logging and policy enforcement, so auditors can trace every finding back to its source document. Live at yanipro.ai.

ActPilot – AI Browser Agent | *Chrome Extension, Azure OpenAI, Groq* Aug 2026

- Built a Chrome/Edge extension (Manifest V3) that reads a page's visible text and a screenshot, then performs real actions on the page (fill fields, click, open links) from natural-language requests.
- Designed the backend to return one structured JSON response per request, limited to a closed set of actions on elements the model was actually given, preventing hallucinated targets.
- Enforced a hard safety rule: the agent fills forms but never submits, pays, deletes or sends anything without explicit instruction.
- Runs on Azure OpenAI's vision model with automatic fallback to Groq on rate-limit errors, keeping the assistant responsive under load.
- Added chart generation, branded PDF export of conversations and rate limiting; published on the Microsoft Edge Add-ons store.

Private Knowledge Assistant | *Agentic RAG, Text-to-SQL, Web RAG, Azure OpenAI* Jan 2026 – Oct 2026

- Built a private AI assistant that answers questions from a user's documents, web pages and live databases in one conversation; a tool-calling agent decides per question whether to search documents, query a database, do both, or simply reply.
- Implemented document and web RAG over PDF, DOCX, TXT, images (OCR fallback for scanned pages) and website URLs, with per-user and per-chat isolation enforced at the vector-store layer.
- Built text-to-SQL/NoSQL over PostgreSQL, MySQL, SQL Server, SQLite and MongoDB, using schema-aware RAG with few-shot examples seeded from foreign-key relationships and successful past queries.
- Designed a five-layer, per-engine read-only guard (keyword blocking, forced rollback, statement timeouts, row caps) and Fernet-encrypted credentials, so connected databases can only be read.
- Every answer cites its source (document page, URL or executed query); responses stream over Server-Sent Events, results render as charts, and accounts use JWT authentication. Live at knowledgeassistant-ai.vercel.app.

Private Voice Assistant | *Twilio, Whisper STT, Orpheus TTS* Aug 2025

- Built a runtime-configurable AI voice helpline for any business, reachable by real phone calls (Twilio) or a browser hold-to-talk demo, both sharing one Groq-powered pipeline.
- Reduced the full conversational turn (speech-to-text, LLM, text-to-speech) to 2–2.7 seconds through connection warm-up and cached greetings.
- Implemented automatic caller language detection from speech, so the assistant replies in the caller's language.
- Integrated Twilio through HTTP polling instead of WebSocket media streams so the service runs on serverless infrastructure. Live at voiceassistant-ai.vercel.app.

WellB – AI-Powered Hotel Management System | *RAG, Llama, DeepSeek, OpenAI* Feb 2025

- Built the "WellB" hotel assistant on a RAG pipeline that combines locally deployed Llama and DeepSeek models with OpenAI GPT.
- Enabled natural-language booking management, including edits with conflict detection and automated rescheduling (DeepSeek for multi-step reasoning).
- Built an emergency alert system with real-time notifications and AI-based priority classification.
- Automated contact extraction and creation with a locally hosted Llama model to keep guest data private.

IntelliReport – AI-Powered Report Generation | *Python, NLP, Transformers, FastAPI* Sep 2024

- Built a report-generation system with automated text correction, topic extraction and summarisation using BERT-based transformer models.
- Combined extractive and abstractive summarisation to produce concise per-topic summaries that keep critical information.

- Rendered reports as Markdown and HTML for the frontend, with PDF download.
- Exposed the system as a FastAPI service with batch processing and real-time progress tracking for large documents.

Smart Media Monitoring Application | *Python, PyTorch* Aug 2023

- Led a media monitoring system combining speech-to-text, sentiment analysis, summarisation, topic modelling, facial recognition and OCR in one pipeline.
- Worked with Urdu and English datasets so the system handles both languages.
- Built a text detection and OCR pipeline for PDFs, images and tables in Urdu, English and Arabic.
- Developed facial recognition to identify and track individuals across media content.

Smart Surveillance System | *Python, PyTorch, TensorFlow* Dec 2022

- Led AI modules for security monitoring over live and IP camera streams.
- Real-time object detection: deployed low-latency detection and tracking models.
- ANPR: built automatic number plate recognition for vehicle identification, with multi-language plate support.
- Geofencing: defined and enforced virtual boundaries within camera streams for automated area monitoring.

AAQAB – Satellite Imagery Intelligence | *Python, YOLO, PyTorch* Feb 2022

- Led object detection on high-resolution satellite imagery using YOLO and ResNet for geospatial intelligence.
- Tiled large images by latitude and longitude and distributed processing across GPUs to make inference tractable.
- Used SAHI sliding-window inference to improve detection and counting accuracy on small objects.

Technical Skills

Generative AI & Agents: LLM Orchestration, Agentic Tool-Calling, RAG Pipelines, LangChain, LangGraph, Prompt Engineering, OpenAI SDK, Azure OpenAI, GCP/Vertex AI, AWS Bedrock, Groq, Open-Source LLMs (Llama, Mistral, DeepSeek), On-Prem LLM Hosting
ML & Computer Vision: PyTorch, TensorFlow, Transformers, YOLO, scikit-learn, Keras, OpenCV, AWS SageMaker
Voice & Speech AI: Groq Whisper STT, Azure Speech, Text-to-Speech, SeamlessM4T (Meta), Twilio Voice, Real-time Audio Pipelines
Data & Retrieval: PostgreSQL, MySQL, SQL Server, MongoDB, Text-to-SQL, SQLAlchemy, Alembic Migrations
Full-Stack & DevOps: FastAPI, React, TypeScript, Docker, AWS Lambda, JWT Auth, CI/CD (pytest/vitest), Vercel, Git
Leadership & Practice: Team Leadership, System Architecture, Security Engineering, Mentorship, Test-Driven Development

Education

University of Engineering and Technology (UET) Oct 2015 – Aug 2019
Bachelor of Science in Computer Engineering *Taxila, Pakistan*

Publications

Meter Digit Recognition Via Faster R-CNN – Muhammad Waqar, M. Waris, et al., M. Yousaf. *IEEE International Conference on Robotics and Automation in Industry (ICRAI)*, Oct 2019. DOI: 10.1109/ICRAI47710.2019.8967357.